Example 12.6

When a pair of fair dice is rolled, what are the probabilities of getting the sum

(i) 7 (ii) 7 or 9 (iii) 7 or 12?

Solution

The sample space $S = \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$

$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),$
$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),$
$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6),$
$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6),$
$(5,1), (5,2), (5,3), (5,4), (5,5), (5,6),$
$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

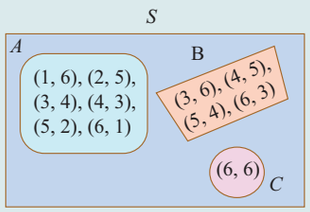
Number of possible outcomes $= 6^2 = 36 = n(S)$

Let A be the event of getting sum 7, B be the event of getting the sum 9 and C be the event of getting sum 12. Then

$A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\} \Rightarrow n(A) = 6$

$B = \{(3,6), (4,5), (5,4), (6,3)\} \Rightarrow n(B) = 4$

$C = \{(6,6)\} \Rightarrow n(C) = 1$



(i) P (getting sum 7) $= P(A)$

$$= \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

(ii) P (getting sum 7 or 9) $= P(A \text{ or } B) = P(A \cup B)$

$$= P(A) + P(B)$$

(Since A and B are mutually exclusive that is, $A \cap B = \varnothing$)

$$= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} = \frac{6}{36} + \frac{4}{36} = \frac{5}{18}$$

(iii) P (getting sum 7 or 12) $= P(A \text{ or } C) = P(A \cup C)$

$$= P(A) + P(C) \quad \text{(since } A \text{ and } C \text{ are mutually exclusive)}$$

$$= \frac{n(A)}{n(S)} + \frac{n(C)}{n(S)} = \frac{6}{36} + \frac{1}{36} = \frac{7}{36}.$$

Example 12.7

Three candidates X, Y, and Z are going to play in a chess competition to win FIDE (World Chess Federation) cup this year. X is thrice as likely to win as Y and Y is twice as likely as to win Z. Find the respective probability of X, Y and Z to win the cup.



Solution

Let A, B, C be the event of winning FIDE cup respectively by X, Y, and Z this year.

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